Yueke Zhang

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Research Focus

My research develops trustworthy and efficient AI for software systems by connecting software engineering, security, and neuroscience. I study how human cognition can guide AI learning and adaptation, how program analysis and formal reasoning can improve reliability, and how contribution accountability can protect open-source ecosystems as AI agents become contributors. My long-term goal is to build agents that learn from experience, improve their capabilities, and carry out complex technical tasks while preserving user intent and human control.

Education

Vanderbilt University, Ph.D. in Computer Science 2023 - 2027 expected
Ph.D. Candidate; Advisor: Yu Huang

Southwest University, Bachelor in Management Information System 2019 - 2023

Publications

[1] **Yueke Zhang**, Zihan Fang, Yuxin Hu, Kevin Leach, Yu Huang: *Unclaimed Territory: Analyzing Subdomain Takeover Risks in GitHub Pages.* **(ICSE 2027).**

[2] **Yueke Zhang**, Yifan Zhang, Zihan Fang, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *ScanCoder: Leveraging Human Attention Patterns to Enhance LLMs for Code.* Proceedings of the ACM on Software Engineering **(FSE 2026).** [Link]

[3] **Yueke Zhang**, Yifan Zhang, Pamela Wisniewski, Fengwei Zhang, Kevin Leach, Yu Huang: *Understanding and Detecting GitHub Impersonation through Automated Authorship Attribution.* ACM Transactions on Software Engineering and Methodology **(TOSEM 2026).** [Link]

[4] **Yueke Zhang**, Anda Liang, Xiaohan Wang, Pamela Wisniewski, Fengwei Zhang, Kevin Leach, Yu Huang: *Who's Pushing the Code? An Exploration of GitHub Impersonation.* IEEE/ACM International Conference on Software Engineering **(ICSE 2025).** [Link]

[5] **Yueke Zhang**, Kevin Leach, Yu Huang: *Leveraging Evidence Theory to Improve Fault Localization: An Exploratory Study.* ACM/IEEE International Symposium on Empirical Software Engineering and Measurement **(ESEM 2023).** [Link]

[6] **Yueke Zhang**, Zihan Fang, Kevin Leach, Yu Huang: *Locating Software Vulnerabilities With Static Analyzers: How Far Are We?* International Conference on Evaluation and Assessment in Software Engineering **(EASE 2026).** [Link]

[7] **Yueke Zhang**, Zihan Fang, Kevin Leach: *Stop the Retrieval Thrash: Brain-Guided Episodic Memory for Repository-Scale Coding Agents*. FSE Companion, HumanAISE Workshop **(2026).** [Link]

[8] **Yueke Zhang**, Zihan Fang, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *CodeACT-R: A Cognitive Simulation Framework for Human Attention in Code Reading.* IEEE/ACM International Conference on Automated Software Engineering, NIER Track **(ASE 2025).** [Link]

[9] **Yueke Zhang**, Yu Huang: *Leveraging Fuzzy System to Reduce Uncertainty of Decision Making in Software Engineering Automation.* Genetic Improvement Workshop at GECCO **(GI 2022).** [Link]

[10] Yifan Zhang, Chen Huang, **Yueke Zhang**, Jiahao Zhang, Toby Jia-Jun Li, Collin McMillan, Kevin Leach, Yu Huang: *EyeMulator: Improving Code Language Models by Mimicking Human Visual Attention.* Annual Meeting of the Association for Computational Linguistics **(ACL 2026).** [Link]

[11] Zihan Fang, **Yueke Zhang**, Thomas Zimmermann, Denae Ford, Yu Huang: *Contribution Patterns in Open Source Software for Social Good: Dynamics, Individuals, and Impact.* Proceedings of the ACM on Human-Computer Interaction **(CSCW 2026).** [Link]

[12] Yifan Zhang, Chen Huang, **Yueke Zhang**, Huajie Shao, Kevin Leach, Yu Huang: *Pre-Training Representations of Binary Code Using Contrastive Learning.* Transactions on Machine Learning Research **(TMLR 2025).** [Link]

[13] Manish Acharya, Zhenyu Liao, **Yueke Zhang**, Kevin Leach, Yu Huang, Yifan Zhang: *VERITAS: Verifier-Guided Proof Search for Zero-Shot Formal Theorem Proving.* AI for Math Workshop at ICML **(AI4Math 2026).** [Link]

Preprints and Manuscripts Under Review

[1] **Yueke Zhang**, Zihan Fang, Kevin Leach, Yu Huang: *GraphAlignCoder: Aligning Program and Proof Graphs for Code Generation.* Under review. [arXiv]

[2] **Yueke Zhang**, Yifan Zhang, Zihan Fang, Kevin Leach, Wei Zhang, Yu Huang: *SCOPE: Leveraging Subgoal Critiques for Code Generation.* Under review. [arXiv]

[3] **Yueke Zhang**, Zihan Fang, Kevin Leach, Yu Huang: *CogCoder: Cognitive Supervision for Code Language Models.* Under review.

[4] **Yueke Zhang**, Yifan Zhang, Kevin Leach, Yu Huang: *CodeGrad: Integrating Multi-Step Verification with Gradient-Based LLM Refinement.* Under review. [arXiv]

[5] Zihan Fang, Yifan Zhang, **Yueke Zhang**, Kevin Leach, Yu Huang: *DPO-F+: Aligning Code Repair Feedback with Developers' Preferences.* Under review. [arXiv]

[6] Zihan Fang, **Yueke Zhang**, Ningzhi Tang, Collin McMillan, Toby Jia-Jun Li, Yu Huang: *From Conversation to Contribution: Characterizing Coding Agent in Open-Source Software.* Under review. [arXiv]

[7] Zihan Fang, **Yueke Zhang**, Yu Huang: *AgentForge: An Immersive Role-Playing Platform for Learning Agentic Software Engineering.* Under review. [arXiv]

[8] Yifan Zhang, Xiaohan Wang, **Yueke Zhang**, Yu Huang, Kevin Leach: *Constraint-Guided Multi-Agent Decompilation for Executable Binary Recovery.* Preprint, 2026. [arXiv]

Research Experience

Research Assistant, Vanderbilt University 2023 - Present

- Conduct research on trustworthy code intelligence, spanning cognitive supervision, code-model training, verification-guided generation and repair, authorship attribution, and empirical software security.

Research Mentoring

Anda Liang, Undergraduate Researcher Fall 2023 - Summer 2024

Co-authored an ICSE 2025 paper on GitHub impersonation; later joined NASA Jet Propulsion Laboratory.

Teaching Experience

Vanderbilt University

- CS8395 Special Topics: Advanced Topics in Software Engineering
- CS-5278 Principles of Software Engineering

Technical Skills

Programming and ML: Python, Java, Shell, PyTorch, TensorFlow. **Methods:** Supervised fine-tuning, preference optimization, reinforcement learning, LoRA/QLoRA, Lean, program graphs, static analysis, and empirical software engineering.
Languages: Mandarin (native), English (fluent).